A Survey of Schema-Based Matching Approaches*

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Abstract. Schema and ontology matching is a critical problem in many application domains, such as semantic web, schema/ontology integration, data warehouses, e-commerce, etc. Many different matching solutions have been proposed so far. In this paper we present a new classification of schema-based matching techniques that builds on the top of state of the art in both schema and ontology matching. Some innovations are in introducing new criteria which are based on (i) general properties of matching techniques, (ii) interpretation of input information, and (iii) the kind of input information. In particular, we distinguish between approximate and exact techniques at schema-level; and syntactic, semantic, and external techniques at element- and structure-level. Based on the classification proposed we overview some of the recent schema/ontology matching systems pointing which part of the solution space they cover. The proposed classification provides a common conceptual basis, and, hence, can be used for comparing different existing schema/ontology matching techniques and systems as well as for designing new ones, taking advantages of state of the art solutions.

1 Introduction

Matching is a critical operation in many application domains, such as semantic web, schema/ontology integration, data warehouses, e-commerce, query mediation, etc. It takes as input two schemas/ontologies, each consisting of a set of discrete entities (e.g., tables, XML elements, classes, properties, rules, predicates), and determines as output the relationships (e.g., equivalence, subsumption) holding between these entities.

Many diverse solutions to the matching problem have been proposed so far, e.g., [2, 5, 21, 23, 40, 46, 49, 50, 52, 57, 76]. Good surveys through the recent years are provided in [39, 62, 75]; while the major contributions of the last decades are presented in [3, 41, 42, 66]. The survey of [39] focuses on current state of the art in ontology matching. Authors review recent approaches, techniques and tools. The survey of [75] concentrates on approaches to ontology-based information integration and discusses general matching approaches that are used in information integration systems. However, none of the above mentioned surveys provide a comparative review of the existing ontology matching techniques and systems. On the contrary, the survey of [62] is devoted to a classification of database schema matching approaches and a comparative

* For more information on the topic (e.g., tutorials, relevant events), please visit the Ontology Matching web-site at www.OntologyMatching.org

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review of matching systems. Notice that these three surveys address the matching problem from different perspectives (artificial intelligence, information systems, databases) and analyze disjoint sets of systems.

This paper aims at considering the above mentioned works together, taking in to account some novel schema/ontology matching approaches, and at providing a common conceptual basis for their analysis. Although, there is a difference between schema and ontology matching problems (see next section for details), we believe that techniques developed for each of them can be of a mutual benefit. Thus, we bring together and discuss systematically recent approaches and systems developed in schema and ontology matching domains. We present a new classification of schema/ontology matching techniques that builds on the work of [62] augmented in [29, 67] and [28]. Some innovations are in introducing new criteria which are based on (i) general properties of matching techniques, (ii) interpretation of input information, and (iii) the kind of input information. In particular, we distinguish between approximate and exact techniques at schema-level; and syntactic, external, and semantic techniques at element- and structure-level. Based on the classification proposed we provide a comparative review of the recent schema/ontology matching systems pointing which part of the solution space they cover. In this paper we focus only on schema-based solutions, i.e., matching systems exploiting only schema-level information, not instance data.

The rest of the paper is organized as follows. Section 2 provides, via an example, the basic motivations and definitions to the schema/ontology matching problem. Section 3 discusses possible matching dimensions. Section 4 introduces a classification of elementary automatic schema-based matching techniques and discusses in detail possible alternatives. Section 5 provides a vision on classifying matching systems. Section 6 overviews some of the recent schema/ontology matching solutions in light of the classification proposed pointing which part of the solution space they cover. Section 7 reports some conclusions and discusses the future work.

2 The Matching Problem

2.1 Motivating Example

To motivate the matching problem, let us use two simple XML schemas that are shown in Figure 1 and exemplify one of the possible situations which arise, for example, when resolving a schema integration task.

Suppose an e-commerce company needs to finalize a corporate acquisition of another company. To complete the acquisition we have to integrate databases of the two companies. The documents of both companies are stored according to XML schemas S and S' respectively. Numbers in boxes are the unique identifiers of the XML elements. A first step in integrating the schemas is to identify candidates to be merged or to have taxonomic relationships under an integrated schema. This step refers to a process of schema matching. For example, the elements with labels Office_Products in S and in S' are the candidates to be merged, while the element with label Digital_Cameras in S' should be subsumed by the element with label Photo_and_Cameras in S. Once the correspondences between two schemas have been determined, the next step has to generate